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Thermal-induced buckling and postbuckling analysis of continuous railway tracks

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ABSTRACT

Railway tracks are vulnerable to buckling when subjected to temperature increases, as occurs frequently during heatwaves. This paper presents an investigation of the lateral buckling and postbuckling response of railway tracks under thermal loading that includes the restraining effect of the ballast in the longitudinal and lateral directions. The principle of stationary total potential is used to develop the differential equations of equilibrium, as well as the bifurcation buckling from this equilibrium, and these equations are shown to be highly non-linear. The buckling analysis shows that first-order symmetric and antisymmetric modes may govern the buckling, and that for very long members the two buckling loads are the same. The highly non-linear equations for the postbuckling response are solved by making recourse to a shooting technique, and it is shown that following the bifurcation buckling under displacement control the response is initially unstable, followed by a stable branch of the postbuckling equilibrium response. Under thermal loading, which occurs in practice, the response is one of snap-through buckling to the remote stable path, and the solution is shown to capture the localisation of the buckling modes that is observed in practice. The effects of the lateral and longitudinal ballast resistance are quantified, and it is shown that the lateral resistance is predominant when compared with the longitudinal resistance. The influence of lateral track imperfections is also studied, which is shown to have considerable effect on the critical temperature.

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1. Introduction

Modern high-speed railways utilise continuously welded rails (CWRs), which are preferred over jointed rails for a number of reasons related to safety, comfort and maintenance costs (Esveld, 2001). The continuity of the steel rails induces compression in the rails because of restraint against thermal expansion, with the potential for subsequent lateral buckling of the rails. These incidences of buckling are observed worldwide, they are on the increase with increasing spells of hot weather, and are an extreme hazard to derailment incidents (Kish and Samavedam, 2013). This paper concerns the analysis of such buckling in railway tracks.

In a broader perspective, the thermal buckling of railway tracks bears some similarities with the thermal upheaval buckling of subsea pipelines (Hobbs, 1981; Ju and Kyriakides, 1988) and of pavements (Kerr and Shade, 1984; Bradford, 2013), as well as the lateral buckling of pipelines (Kerr, 1971; Peek and Yun, 2007). It is noted that this general area of unilateral buckling (Smith et al., 2000) finds widespread application in many facets of engineering mechanics.

There are several investigations of thermal-induced railway track buckling to be found in the literature, with varying degrees of sophistication. In a series of papers, Kerr (Kerr, 1978; Kerr, 1980; Donley and Kerr, 1987) has provided critical reviews of earlier work, refined models of track ballast restraint and formulated the so-called "safe temperature", below which it is expected that lateral buckling will not occur. Kish et al. (1982) provided experimental results obtained from two tests, one on a straight track and one on a curved one, but the full postbuckling curve could not be traced as a branch of it is unstable and was not able to be controlled. Parametric studies of the stability of railway tracks were also conducted by Samavedam et al. (1983, 1993), considering such factors as the axial resistance, lateral resistance and track misalignments. All of these research contributions focus on the safe temperature, and very few research investigations have addressed the entire postbuckling response of railway tracks under thermal loading when compared with determining the safe temperature.

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In more recent times, the finite element method has been used to investigate the response of railway tracks under thermal loading. A three-dimensional finite element CWR model was developed by Lim et al. (2008), and used subsequently to undertake an extensive buckling analysis with the conclusion that the buckling is a three-dimensional problem. Later, parametric studies of the influence of the track ballast and track irregularity on the stability of CWRs were undertaken by Lim et al. (Van, 1997) using this finite element model. Van (Hetenyi, 1974) proposed a finite element solution for the stability of CWR tracks and a bespoke code was developed.

The track buckling problem can be further simplified as being a beam on an elastic foundation, for which extensive research has been published (Timoshenko and Gere, 1961; Wu and Zhong, 1999; Song and Li, 2007; Wang, 2010). Li and Batra (2007) analysed the deformations of pinned-pinned and fixed-fixed Euler-Bernoulli beams supported on non-linear elastic foundations and heated uniformly into the postbuckling regime, and obtained quasi-periodic mode shapes for a beam of finite length. Most research on modelling railway tracks on a foundation (to model the ballast) has been based on a track of finite length, with buckling being confined to a small region by imposing additional constraints that do not in fact exist. In reality, railway tracks are extremely long compared with the buckling lengths observed, and these buckles most often show a localisation of the mode and are not periodic or harmonic. Several researchers have investigated buckling localisation. Wadee et al. (1997) developed a static description of this static problem which has analogies to certain aspects of dynamic theory, in that both kinetic and potential energy components were identified, and solved the problem using an asymptotic perturbation solution around loads close to the buckling load. Chen and Baker (2003) proposed a Rayleigh-Ritz procedure for the localised buckling of a strut on a non-linear foundation by expanding the deflected shape of the strut into a series of Hermite orthogonal functions that are energy-integrable in an infinite region. Because these representations do not consider antisymmetric buckling, Yang and Bradford (2015) extended the model of Wadee et al. (1997), accounting for both symmetric and antisymmetric buckling and in the context of railway buckling.

This paper considers the buckling and postbuckling of railway tracks caused by thermal effects typical of those encountered in prolonged heatwaves. Both the lateral and longitudinal restraint provided by the ballast are modelled using a mechanics-based quasi-empirical model that is appropriate for elastic analysis as the temperature is assumed to increase monotonically, and symmetric and antisymmetric modes are admissible in the formulation. It is shown that once a critical temperature is reached, the postbuckling equilibrium path is initially unstable and then stable, with a localisation of the buckling mode taking place as the path is followed. It is shown that under "temperature control" the track will snap through to an adjacent equilibrium accompanied by a significant decrease of the axial compression and a profound localisation of the mode shape, which is consistent with field observations.

2. Geometric nonlinearity

2.1. Basic assumptions

The formulation in this study is based on the following considerations and assumptions: (1) Euler-Bernoulli bending theory is adopted so that the cross-section is perpendicular to the axis of the beam in the deformed configuration and local buckling is not considered, nor are distortion or warping of the rails; (2) the strain is small, and two axis systems are used to describe the configuration of the rail as shown in Fig. 1. The first is *OXY*, being a right-handed rectangular coordinate system that is fixed in space

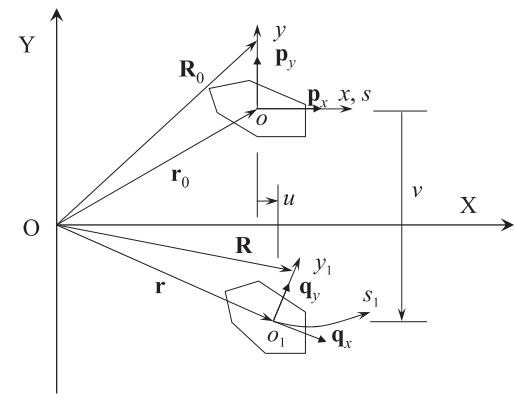


Fig. 1. Coordinate system and position vectors.

as a Lagrangean spatial reference, while the second axis system *osy* is a right-handed curvilinear orthogonal axis system, and is the material-level description of the system. In its undeformed configuration, this axis system is a rectangular orthogonal system *oxy* in which *o* is the centroid of the cross-section, *ox* is perpendicular to the cross-section and *oy* the minor principal axis of the cross-section. In the reference space, *OX* and *OY* are parallel to *ox* and *oy*, and unit vectors fixed in the reference description *OXY* are $\mathbf{p}_x$ and $\mathbf{p}_y$ in the directions of *ox* and *oy* respectively.

In the deformed configuration, the centroid of the cross-section *o* undergoes a displacement (u, v) to the position o_1, and the straight axis *ox* deforms to a curve o_1s_1, so the original material description system *oxy* changes to a new system $o_1s_1y_1$. A unit vector $\mathbf{q}_x$ is defined along the tangent direction of the axis o_1s_1 and which remains normal to the cross-section of the rail, whilst $\mathbf{q}_y$ is the unit vector in the direction of the axis o_1y_1 and which is perpendicular to $\mathbf{q}_x$ and coincident with the minor principal axis of the cross-section in the deformed configuration. The axis system $o_1s_1y_1$ and unit vectors $\mathbf{q}_x$ and $\mathbf{q}_y$ move and rotate with the deformation of the rail.

2.2. Rotation and curvature

The original position vector of the centroid *o* of the axis system *oxy* in the reference space *OXY* is defined by $\mathbf{r}_0$, so that the unit vector $\mathbf{p}_x$ in the undeformed configuration is given by Pi et al. (2007)

$$\mathbf{p}_x = \frac{\mathrm{d}\mathbf{r}_0}{\mathrm{d}x}. \tag{1}$$

When the rail deforms, the position vector of the centroid o_1 of the material description system $o_1x_1y_1$ in the spatial reference system *OXY* is denoted as $\mathbf{r}$. The unit vector $\mathbf{q}_x$ in the deformed configuration is therefore

$$\mathbf{q}_x = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}s_1} = \frac{1}{1+\varepsilon}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}x}, \tag{2}$$

in which $\mathrm{d}s_1 = (1+\varepsilon)\mathrm{d}x$ is the arc coordinate in the deformed configuration, with ε being the elongation per unit length, *i.e.* the strain in the tangent direction of s_1.

The position vector $\mathbf{r}$ in the deformed configuration can be expressed in terms of the position vector $\mathbf{r}_0$ and the unit vectors $\mathbf{p}_x$ and $\mathbf{p}_y$ in the undeformed configuration as

$$\mathbf{r} = \mathbf{r}_0 + u\mathbf{p}_x + v\mathbf{p}_y, \tag{3}$$

and differentiating both sides of this equation with respect to the arc length coordinate s_1 as well as using Eqs. (1) and (2) produces

$$\mathbf{q}_x = \frac{1}{1+\varepsilon}\left(\mathbf{p}_x + u'\mathbf{p}_x + v'\mathbf{p}_y\right), \tag{4}$$

where $()' = \mathrm{d}()/\mathrm{d}x$.

The angle between the tangential direction of the arc length coordinate s_1 and the axis ox is denoted as θ, so the unit vector $\mathbf{q}_x$ can also be expressed in terms of $\mathbf{p}_x$, $\mathbf{p}_y$ and θ as

$$\mathbf{q}_x = \cos\theta\,\mathbf{p}_x + \sin\theta\,\mathbf{p}_y. \tag{5}$$

The curvature of the member in the deformed configuration can be written in terms of the derivatives of the unit vectors as (Simo and Vu-Quoc, 1991; Lai et al., 2010)

$$\frac{\mathrm{d}\mathbf{q}_y}{\mathrm{d}s_1} = \kappa\mathbf{q}_x, \quad \frac{\mathrm{d}\mathbf{q}_x}{\mathrm{d}s_1} = -\kappa\mathbf{q}_y, \tag{6}$$

where κ is the curvature of the rail in the direction of the unit vector $\mathbf{q}_x$. Eq. (6) can be written in matrix form as

$$\begin{Bmatrix} \mathrm{d}\mathbf{q}_x/\mathrm{d}s_1 \\ \mathrm{d}\mathbf{q}_y/\mathrm{d}s_1 \end{Bmatrix} = \begin{bmatrix} 0 & -\kappa \\ \kappa & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{q}_x \\ \mathbf{q}_y \end{Bmatrix} \tag{7}$$

and the unit vectors $\mathbf{q}_x$ and $\mathbf{q}_y$ as

$$\begin{Bmatrix} \mathbf{q}_x \\ \mathbf{q}_y \end{Bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{Bmatrix} \mathbf{p}_x \\ \mathbf{p}_y \end{Bmatrix}. \tag{8}$$

Equating Eqs. (4) and (5) leads to

$$\cos\theta = \frac{1+u'}{1+\varepsilon}, \quad \sin\theta = \frac{v'}{1+\varepsilon} \tag{9}$$

and substituting this into Eq. (8) leads to

$$\begin{Bmatrix} \mathbf{q}_x \\ \mathbf{q}_y \end{Bmatrix} = \mathbf{T} \begin{Bmatrix} \mathbf{p}_x \\ \mathbf{p}_y \end{Bmatrix}, \tag{10}$$

in which the rotation matrix $\mathbf{T}$ is defined by

$$\mathbf{T} = \frac{1}{1+\varepsilon} \begin{bmatrix} 1+u' & v' \\ -v' & 1+u' \end{bmatrix}. \tag{11}$$

Differentiating both sides of Eq. (10) with respect to x produces

$$\begin{Bmatrix} \mathrm{d}\mathbf{q}_x/\mathrm{d}x \\ \mathrm{d}\mathbf{q}_y/\mathrm{d}x \end{Bmatrix} = \frac{\mathrm{d}\mathbf{T}}{\mathrm{d}x} \begin{Bmatrix} \mathbf{p}_x \\ \mathbf{p}_y \end{Bmatrix}, \tag{12}$$

and substituting Eqs. (7) and (10) into Eq. (12) leads to a relationship between the strain ε, curvature κ and rotation matrix $\mathbf{T}$ as

$$(1+\varepsilon) \begin{bmatrix} 0 & -\kappa \\ \kappa & 0 \end{bmatrix} = \frac{\mathrm{d}\mathbf{T}}{\mathrm{d}x}\mathbf{T}^{-1}. \tag{13}$$

It can be deduced from Eqs. (9) and (11) that the rotation matrix satisfies the orthogonality condition

$$\mathbf{T}\mathbf{T}^{\mathrm{T}} = \mathbf{I} \tag{14}$$

in which $\mathbf{I}$ is the identity matrix, and so

$$\mathbf{T}^{-1} = \mathbf{T}^{\mathrm{T}}. \tag{15}$$

Because $\mathbf{q}_x$ is a unit vector, $|\mathbf{q}_x| = 1$ and so from Eq. (4),

$$1+\varepsilon = \sqrt{v'^2 + (1+u')^2}. \tag{16}$$

Therefore, the curvature κ in the deformed state can be deduced from Eq. (13) as

$$\kappa = \frac{u''v' - v''(1+u')}{\left[v'^2 + (1+u')^2\right]^{3/2}}. \tag{17}$$

2.3. Position and finite strains

With respect to the vector $\mathbf{R}$ in Fig. 1, the deformation gradient tensor $\mathbf{F}$ in the deformed configuration can be written as (Lai et al., 2010)

$$\mathbf{F} = \left\langle \frac{\partial\mathbf{R}}{\partial x}, \frac{\partial\mathbf{R}}{\partial y} \right\rangle^{\mathrm{T}} \tag{18}$$

and the Green–Lagrange strain tensor can be expressed in terms of the deformation gradient tensor as

$$\mathbf{E} = \begin{bmatrix} \varepsilon_{xx} & \gamma_{xy}/2 \\ \gamma_{yx}/2 & \varepsilon_{yy} \end{bmatrix} = \tfrac{1}{2}\left(\mathbf{F}^{\mathrm{T}}\mathbf{F} - \mathbf{I}\right), \tag{19}$$

from which the normal strain ε_{xx} can be obtained as (Love, 1944)

$$\varepsilon_{xx} = \frac{1}{2}\left(\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}x}\cdot\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}x} - 1\right) = (1+\varepsilon)^2\frac{1}{2}\left(\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}s_1}\cdot\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}s_1} - 1\right), \tag{20}$$

which leads to

$$\varepsilon_{yy} = 0,\ \gamma_{xy} = \gamma_{yx} = 0,\ \varepsilon_{xx} = \frac{1}{2}\left[(1+\varepsilon)^2(1+y\kappa)^2 - 1\right]. \tag{21}$$

By substituting the expressions for the curvature κ and strain ε, the normal strain can be formulated as

$$\varepsilon_{xx} = \frac{1}{2}\left\{ v'^2 + u'^2 + 2u' + \frac{2y\left[u''v' - v''(1+u')\right]}{\sqrt{v'^2 + (1+u')^2}} + \frac{y^2\left[u''v' - v''(1+u')\right]^2}{\left[v'^2 + (1+u')^2\right]^2} \right\}, \tag{22}$$

from which the axial strain at the centroid of the cross-section can be expressed as

$$\varepsilon_{xx|y=0} = u' + \tfrac{1}{2}u'^2 + \tfrac{1}{2}v'^2. \tag{23}$$

2.4. Thermal strain

The thermal strain is assumed to be uniform through the cross-section of the rail and along its length. The thermal Green-Lagrange strain at the centroid can then be obtained from Eq. (21) as

$$\varepsilon_{xx}^{T} = \frac{1}{2}\left[(1+\alpha T)^2 - 1\right] = \alpha T + \tfrac{1}{2}(\alpha T)^2 \approx \alpha T \tag{24}$$

since in reality $(\alpha T)^2 <<< \alpha T$, in which T is the temperature increase above the neutral temperature and α the coefficient of thermal expansion.

3. Analytical formulation

The analysis in this investigation is based on the following assumptions: (1) the track is assumed to remain elastic in the post-buckling stage and (2) only one rail is considered in the analysis. In reality, two rails are connected by sleepers, and in the event of lateral buckling the deformations of the two rails are considered to be the same and their buckled configurations are parallel, and so one rail and the corresponding lateral and longitudinal resistances only are considered in the analysis.

3.1. Non-linear equilibrium

The total potential Π of the railway track, including the contribution of its bending stiffness in its deformed configuration, its axial stiffness and the lateral and longitudinal resistance of the

ballast can be written as

$$\Pi = \int_{-\infty}^{\infty} \tfrac{1}{2} EI\kappa^2 \mathrm{d}x + \int_{-\infty}^{\infty} \tfrac{1}{2} AE\left(\varepsilon_{xx}|_{y=0} - \varepsilon_{xx}^T\right)^2 \mathrm{d}x + \int_{-\infty}^{\infty}\int_0^u F_1(u)\mathrm{d}u\mathrm{d}x + \int_{-\infty}^{\infty}\int_0^v F_2(v)\mathrm{d}v\mathrm{d}x, \tag{25}$$

in which $F_1(u)$ is the longitudinal resistance due to the interaction of the track and the ballast which is a function of the longitudinal displacement u, and $F_2(v)$ the lateral resistance of the ballast, which is composed of the contribution of the sleeper base, crib ballast and shoulder ballast (Le Pen and Powrie, 2011), and it is a function of the lateral displacement v.

The principle of stationary total potential enables the postbuckling equilibrium state to be determined from

$$\delta\Pi = 0 \tag{26}$$

with respect to any admissible infinitesimal variations of the deformations. By substituting Eqs. (17), (23) and (24) into Eq. (25) and then Eq. (26), the non-linear differential equation of equilibrium is

$$\frac{EI}{\Gamma^5}\left(\Phi_1\Gamma^2 u^{\mathrm{iv}} + \Phi_2\Gamma^2 v^{\mathrm{iv}} + \Phi_3\Gamma u''' + \Phi_4\Gamma v''' + \Phi_5\right) + AE\Phi_6 + F_1(u) = 0 \tag{27}$$

in the longitudinal (x - axis) direction, and

$$\frac{EI}{\Gamma^5}\left(\Lambda_1\Gamma^2 u^{\mathrm{iv}} + \Lambda_2\Gamma^2 v^{\mathrm{iv}} + \Lambda_3 u''' + \Lambda_4\Gamma v''' + \Lambda_5\Gamma\right) + AE\Lambda_6 + F_2(v) = 0 \tag{28}$$

in the lateral (y - axis) direction, in which

$$\Gamma = u'^2 + v'^2 + 2u' + 1 \tag{29}$$

and the terms $\Phi_1, .., \Phi_6, \Lambda_1, .., \Lambda_6$ are given in the Appendix.

Because railway tracks are usually long, the regularity boundary conditions that

$$u(\pm l) = 0,\ v(\pm l) = 0,\ v'(\pm l) = 0;\ l \to \infty \tag{30}$$

may be sensibly hypothesised.

3.2. Buckling analysis

The critical condition for buckling can be stated as being that the second variation of the total potential of the track system vanishes for any admissible infinitesimal variations of the deformations, that is

$$\delta^2\Pi = 0. \tag{31}$$

Because the buckling is a bifurcation of equilibrium, the prebuckled and the buckled configurations intersect at the same equilibrium point. By substituting the trivial equilibrium configuration that $u \equiv 0$ and $v \equiv 0$ into Eq. (31) and using Eq. (25), the governing differential equation for buckling can be obtained as

$$EIv_b^{\mathrm{iv}} + AE\alpha T v_b'' + (\mathrm{d}F_2/\mathrm{d}v)_{v=0} v_b = 0, \tag{32}$$

in which $v_b = \delta v$ is the buckling deformation. Again using the regularity conditions for the buckling deformations that

$$v_b(\pm l) = 0,\ v_b'(\pm l) = 0;\ l \to \infty \tag{33}$$

leads to the general solution of Eq. (32) as

$$v_b = C_1\cos\mu_1 x + C_2\sin\mu_1 x + C_3\cos\mu_2 x + C_4\sin\mu_2 x, \tag{34}$$

in which

$$\mu_1 = \sqrt{\frac{\tau}{2} - \sqrt{\frac{\tau^2}{4} - k}}, \quad \mu_2 = \sqrt{\frac{\tau}{2} + \sqrt{\frac{\tau^2}{4} - k}} \tag{35}$$

with

$$\tau = \frac{\alpha T}{r^2},\ k = \frac{(\mathrm{d}F_2/\mathrm{d}v)_{v=0}}{EI},\ r = \sqrt{\frac{I}{A}} \tag{36}$$

and the constants of integration $C_1, .., C_4$ satisfying

$$\begin{bmatrix} \cos(\mu_1 l) & \sin(\mu_1 l) & \cos(\mu_2 l) & \sin(\mu_2 l) \\ \cos(\mu_1 l) & -\sin(\mu_1 l) & \cos(\mu_2 l) & \sin(\mu_2 l) \\ -\mu_1\cos(\mu_1 l) & \mu_1\sin(\mu_1 l) & -\mu_2\cos(\mu_2 l) & \mu_2\sin(\mu_2 l) \\ \mu_1\cos(\mu_1 l) & \mu_1\sin(\mu_1 l) & \mu_2\cos(\mu_2 l) & \mu_2\sin(\mu_2 l) \end{bmatrix} \times \begin{Bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{Bmatrix} = \mathbf{0}. \tag{37}$$

The existence of nontrivial solutions in Eq. (34) requires that the determinant of the coefficient matrix in Eq. (37) vanishes, leading to the characteristic equation

$$[\mu_2\cos(\mu_2 l)\sin(\mu_1 l) - \mu_1\sin(\mu_2 l)\cos(\mu_1 l)] \cdot [\mu_1\cos(\mu_2 l)\sin(\mu_1 l) - \mu_2\sin(\mu_2 l)\cos(\mu_1 l)] = 0. \tag{38}$$

When the first factor in the characteristic Eq. (38) vanishes, the buckling mode is antisymmetric and when the second factor vanishes, the mode is symmetric.

4. Imperfection sensitivity

Bifurcation buckling takes place in perfectly-straight railway tracks at a certain temperature, which is then followed by a nonlinear postbuckling process. If such tracks have an initial geometric imperfection, non-linear deformations will develop from the commencement of the temperature increase, thereby having a different equilibrium path to perfect tracks. The total potential of a track with an initial geometric imperfection v_0 can be written using Eq. (25) as

$$\Pi = \int_{-\infty}^{\infty} \tfrac{1}{2} EI[\kappa(u, v+v_0) - \kappa(0, v_0)]^2 \mathrm{d}x + \int_{-\infty}^{\infty} \tfrac{1}{2} AE\left[\varepsilon_{xx}(u, v+v_0)_{|y=0} - \varepsilon_{xx}(0, v_0)_{|y=0} - \varepsilon_{xx}^T\right]^2 \mathrm{d}x + \int_{-\infty}^{\infty}\int_0^u F_1(u)\mathrm{d}u\mathrm{d}x + \int_{-\infty}^{\infty}\int_0^v F_2(v)\mathrm{d}v\mathrm{d}x\,. \tag{39}$$

Eq. (17) can be simplified conveniently by omitting the contribution of the axial deformation (Pi and Trahair, 1994) to produce

$$\kappa = \frac{v''}{\left(v'^2 + 1\right)^{3/2}}, \tag{40}$$

which when substituted along with Eq. (23) into Eq. (39) yields

$$\Pi = \int_{-\infty}^{\infty} \frac{1}{2} EI\left\{\frac{v'' + v''_0}{\left[(v' + v'_0)^2 + 1\right]^{3/2}} - \frac{v''_0}{\left(v_0'^2 + 1\right)^{3/2}}\right\}^2 \mathrm{d}x + \int_{-\infty}^{\infty}\int_0^u F_1(u)\mathrm{d}u\mathrm{d}x + \int_{-\infty}^{\infty}\int_0^v F_2(v)\mathrm{d}v\mathrm{d}x + \int_{-\infty}^{\infty} \tfrac{1}{2} AE\left[u' + \tfrac{1}{2}u'^2 + \tfrac{1}{2}\left(v' + v'_0\right)^2 - \tfrac{1}{2}v_0'^2 - \alpha T\right]^2 \mathrm{d}x. \tag{41}$$

Invoking the principle of stationary total potential, the equilibrium configuration makes the total potential stationary with respect to any admissible infinitesimal variations of the deformation (*i.e.* Eq. (26)), leading to the differential equation of equilibrium

$$AE\left(-3u''u' - \frac{3}{2}u''u'^2 + \bar{\Phi}_1 u'' + \bar{\Phi}_2 u' + \bar{\Phi}_3\right) + F_1(u) = 0 \tag{42}$$

in the longitudinal direction, and to

$$\frac{EI}{D_1^3 D_2^{10}}\left[D_1^3\left(\bar{\Lambda}_1 v^{\mathrm{iv}} + \bar{\Lambda}_2 v''' + \bar{\Lambda}_3\right) + D_2^3 \bar{\Lambda}_4\right] + AE\bar{\Lambda}_5 + F_2(v) = 0 \tag{43}$$

in the lateral direction, where D_1, D_2, $\bar{\Phi}_1$, $\bar{\Phi}_2$, $\bar{\Phi}_3$, $\bar{\Lambda}_1$, .., $\bar{\Lambda}_5$ are defined in the Appendix.

5. Numerical solution procedure

5.1. Single shooting procedure

The equilibrium Eqs. (27) and (28) for a straight track and (42) and (43) for one with a lateral imperfection can be written as a system of non-linear differential equations in the form

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}x} = \mathbf{f}(x, \mathbf{v}), \tag{44}$$

in which

$$\mathbf{v} = \left\langle v, v', v'', v''', T, u, u'\right\rangle^{\mathrm{T}}. \tag{45}$$

The expression for $\mathbf{f}(x, \mathbf{v})$ can be obtained explicitly from the equilibrium equations and, because it is complicated, it is not given here. The boundary conditions may be stated in matrix format as

$$\mathbf{A}\mathbf{v}(0) + \mathbf{B}\mathbf{v}(l) = \mathbf{C}, \tag{46}$$

in which

$$\mathbf{A} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \end{bmatrix}, \quad \mathbf{C} = \begin{Bmatrix} v(0) \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{Bmatrix} \tag{47}$$

for symmetric buckling, and

$$\mathbf{A} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \end{bmatrix}, \quad \mathbf{C} = \begin{Bmatrix} \cdot \\ \theta(0) \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{Bmatrix} \tag{48}$$

for antisymmetric buckling.

The initial conditions at the point $x=0$ can be expressed as $\mathbf{h} = \langle v(0), 0, h_1, 0, h_2, 0, h_3\rangle^{\mathrm{T}}$ for symmetric buckling, and as $\mathbf{h} = \langle 0, \theta(0), 0, h_1, h_2, 0, h_3\rangle^{\mathrm{T}}$ for antisymmetric buckling, where h_1, h_2 and h_3 are parameters to be determined. By considering these initial values, the boundary value problem given by Eqs. (44) and (45) can be converted to the initial value problem

$$\hat{\mathbf{v}}(x; \mathbf{h}) = \mathbf{h} + \int_0^l \mathbf{f}(x, \mathbf{v})\mathrm{d}x, \tag{49}$$

which can be integrated explicitly using the Runge-Kutta technique, which is of fourth order. Using this, the function ζ can be represented by the expression

$$\zeta_{i+1|\mathbf{h}} = \zeta_{i|\mathbf{h}} + \frac{d}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \quad (i = 1, 2, .., m), \tag{50}$$

in which $\zeta_{i+1|\mathbf{h}} = \hat{\mathbf{v}}(x_i; \mathbf{h})$, x_i are uniform discrete values of the x-coordinate with an interval d, and $\mathbf{k}_1$, .., $\mathbf{k}_4$ are parameters defined by

$$\mathbf{k}_1 = \mathbf{f}(x_i, \zeta_i), \ \mathbf{k}_2 = \mathbf{f}\left(x_i + \frac{d}{2}, \zeta_i + \frac{d\mathbf{k}_1}{2}\right),$$
$$\mathbf{k}_3 = \mathbf{f}\left(x_i + \frac{d}{2}, \zeta_i + \frac{d\mathbf{k}_2}{2}\right), \mathbf{k}_4 = \mathbf{f}(x_i + d, \zeta_i + d\mathbf{k}_3). \tag{51}$$

For a given value $v(0)$ for symmetric buckling or $\theta(0)$ for antisymmetric buckling, the boundary condition at the other end ($x=l$) can be rewritten as

$$\mathbf{A}\mathbf{h} + \mathbf{B}\zeta_{m|\mathbf{h}} = \mathbf{C}, \tag{52}$$

which represents a non-linear equation in the unknown variable $\mathbf{h}$. This equation can be solved using Newton–Raphson iteration to produce the root $\mathbf{h}^*$, so that the postbuckling configuration for the track for a predefined value of $v(0)$ for symmetric buckling or $\theta(0)$ for antisymmetric buckling can obtained as

$$\mathbf{v}(x_i) = \hat{\mathbf{v}}(x_i; \mathbf{h}) \quad (i = 1, 2, ..., m). \tag{53}$$

5.2. Multiple shooting techniques

The configuration at the early stages of postbuckling can be solved using the single shooting method strategy, but at later stages Eq. (52) is difficult to solve. Accordingly, the single shooting method is used to solve Eq. (53) and these results are fed into a multiple shooting method technique to trigger the postbuckling analysis (Osborne, 1969; Keller, 1992). For this, the values of $\mathbf{v}(x_i)$, $i=0, 1, .., m$ are computed simultaneously by iteration (Stoer and Bulirsch, 2002). The exact values of $\mathbf{v}(x_i)$ are denoted as ψ_i, *i.e.*

$$\mathbf{v}(x_i) = \psi_i, \quad (i = 0, 1, 2, .., m), \tag{54}$$

with the function $\mathbf{v}(x)$ being pieced together by the equations

$$\mathbf{v}(x) = \mathbf{v}(x; x_i, \psi_i), \quad (i = 0, 1, 2, .., m-1). \tag{55}$$

Since each piecewise function is continuous at its connection points, *viz.* at the other end of the interval, the values obtained from Eq. (55) using the Runge–Kutta method of Eq. (52) satisfy the conditions

$$\mathbf{v}(x_{i+1}; x_i, \psi_i) = \psi_{i+1}, \quad (i = 0, 1, 2, .., m-1), \tag{56}$$

from which the non-linear system of equations

$$\mathbf{F}(\psi_0, \psi_1, ..., \psi_m) = \left\{ \begin{array}{c} \mathbf{v}(x_1; x_0, \psi_0) - \psi_1 \\ \mathbf{v}(x_2; x_1, \psi_1) - \psi_2 \\ \cdot \\ \cdot \\ \cdot \\ \mathbf{v}(x_m; x_{m-1}, \psi_{m-1}) - \psi_m \\ \mathbf{A}\psi_0 + \mathbf{B}\psi_m - \mathbf{C} \end{array} \right\} = \mathbf{0} \tag{57}$$

is obtained, where the last row is the boundary condition given in Eq. (46). This non-linear equation can be solved using Newton-Raphson iteration at each control parameter v_0 or θ_0, with the entire postbuckling regime being generated by varying the control parameter step by step, with the results for the current step being used as the initial value for the subsequent step.

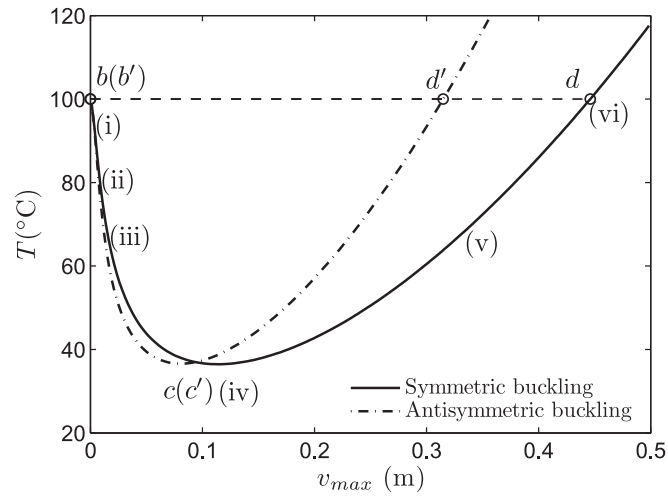


Fig. 2. Variation of maximum lateral displacement with temperature increase.

6. Applications

6.1. Material properties and ballast resistance

The lateral and longitudinal frictional forces between the rail-sleeper system and the ballast act in opposite directions to the displacements. Field tests reported by many investigators (Donley and Kerr, 1987; Dogneton, 1975; Jeong et al., 1986; Samavedam et al., 1986) suggest the forces F_1 and F_2 in Eq. (25) can be represented conventionally as

$$F_1(u) = \rho_1 \tanh(\eta_1 u), \quad F_2(v) = \rho_2 \tanh(\eta_2 v) \tag{58}$$

in the longitudinal and lateral directions respectively. The parameters ρ_1 and ρ_2 have the units of [force/length] and η_1 and η_2 the units of [length^{-1}]. In the current investigation, a 132RE rail is adopted for the analysis with the related parameters, selected from (Donley and Kerr, 1987), being: $A = 16.70 \times 10^{-3}\,\mathrm{m}^2$, $I = 12.14 \times 10^{-6}\,\mathrm{m}^4$, $\rho_1 = \rho_2 = 3920\,\mathrm{N/m}$, $\eta_1 = 200\,\mathrm{m}^{-1}$, $\eta_2 = 300\,\mathrm{m}^{-1}$, $\alpha = 10.0 \times 10^{-6}/°\mathrm{C}$, $E = 210 \times 10^9\,\mathrm{N/m}^2$.

6.2. Buckling and postbuckling behaviour

When the railway track is of finite length l, it can be seen from Eq. (38) that symmetric or antisymmetric buckling will take place as being the lowest buckling mode. It can be shown that the difference between the first symmetric buckling mode and the first antisymmetric buckling load decreases with an increase of the length l, and it can be shown further for an infinitely long track that both modes are the same, *viz.*

$$\tau_{cr} = \lim_{l \to \infty} \tau = 2\sqrt{\frac{\rho_2 \eta_2}{EI}} \tag{59}$$

and the corresponding wavelength at this buckling state is

$$L_{wave}^{cr} = 2\pi \sqrt{\frac{2}{\tau_{cr}}}. \tag{60}$$

The postbuckling responses of the railway track under thermal loading are shown in Figs. 2–8. It can be seen in Fig. 2 that the initial stages of postbuckling are characterised by an unstable stage in which the temperature decreases as the deformation increases from b to c (or from b' to c') and then increases with further a further increase of the deformation to d (or d') and beyond on the stable equilibrium path. In reality, the displacement does not control the phenomenon and so the track will snap through from the buckling point b–d (or b'–d'), producing a dynamic effect that may amplify the deformation. The temperature at the point b in Fig. 2 is defined as the critical temperature T_{cr}, which when reached will cause the railway track to buckle immediately to the remote point d and the deformations will increase subsequently with an increase of the temperature. On the other hand, the temperature at the point c in Fig. 2 is the safe temperature T_{safe}, below which the buckling will not occur. In the range $T_{safe} < T < T_{cr}$, snap-through buckling is possible because of disturbances, and the phenomenon of wheelset hunting in the bogies will produce such a lateral disturbance. The postbuckling deformations are shown in Fig. 3 at the points (i)–(vi) indicated on the equilibrium curve shown in Fig. 2. It can be seen that on traversing the equilibrium path from b to c to d, the postbuckling mode becomes more localised. In conditions of heatwaves, such localised buckling modes are observed widely in practice.

The variation of the maximum rotation $\theta_{\max}$ of the railway track with the temperature increase is shown in Fig. 4, in which it can be seen that the difference between the cases of symmetric and antisymmetric bucking is not large, the latter being slightly larger than the former.

For periodic buckling, the wavelength is that from a point on the buckle to an adjacent point of the same phase, *e.g.* from one crest to an adjacent crest or a zero point to another zero point, whilst for a localised configuration the wavelength is not constant along the track. Fig. 5 shows the maximum wavelength L_{wave} for symmetric buckling and for antisymmetric buckling. Fig. 6 illustrates that the maximum wavelength L_{wave} increases with the evolution of the postbuckling, and when the track snaps through to the point $d(d')$ it is nearly three times the value at the point of buckling. Following the snap through to the point $d(d')$, the wavelength of the buckle increases with an increase of the temperature.

Fig. 7 shows the axial force N in the track as a function of the temperature increase T at the midpoint ($x=0$). Prior to attaining the critical temperature at $b(b')$, the increase in the axial force is linear in accordance with $N = AE\alpha T$. The track then snaps through at buckling to $d(d')$ with a very significant loss of the compression that is built up by thermal loading. It is noteworthy that the postbuckling equilibrium paths in Fig. 7 for symmetric and antisymmetric buckling modes are almost the same.

The relationship between the axial force in the track and the maximum buckling deformation v_{max} is shown in Fig. 8. The points $d(d')$ on this curve are the remote stable equilibrium points, to which the track snaps through, with considerable loss of compression. Following this, the axial force remains fairly constant while the buckling deformations further evolve.

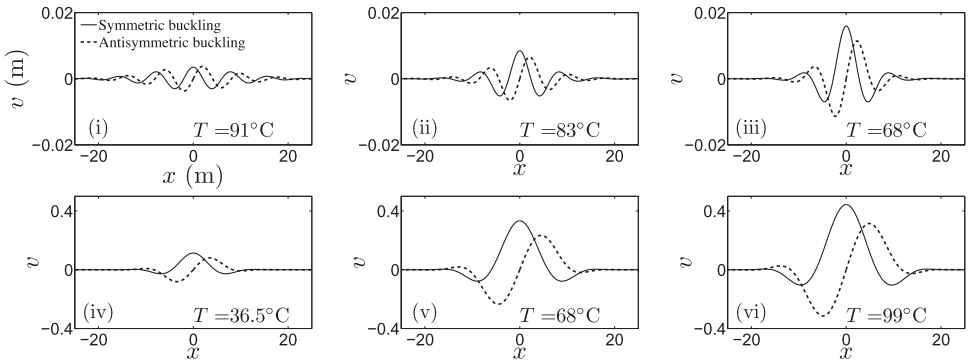


Fig. 3. Deformation shape in the postbuckling stage.

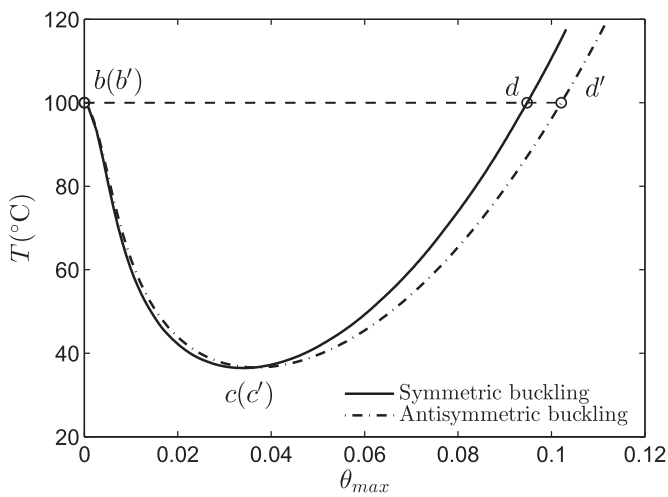


Fig. 4. Variation of maximum rotation with temperature increase.

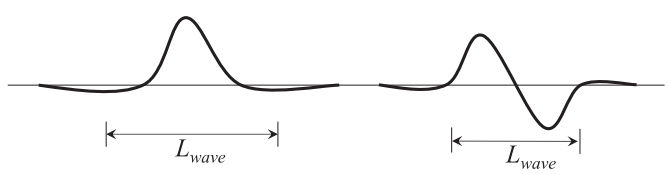


Fig. 5. Maximum wave length (L_{wave}).

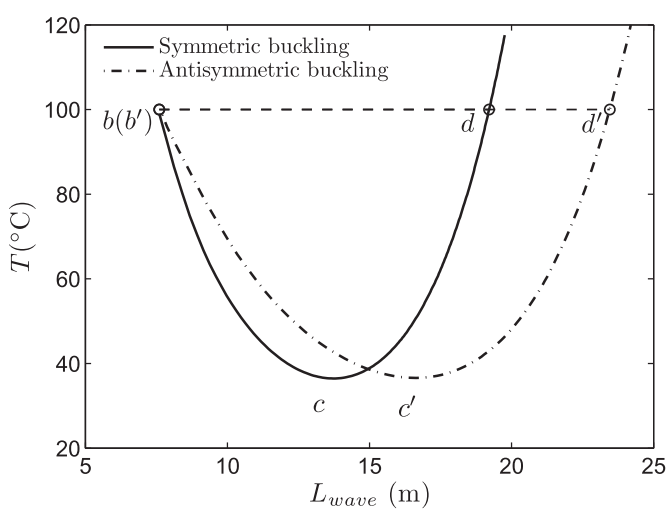


Fig. 6. Variation of maximum wave length with temperature increase.

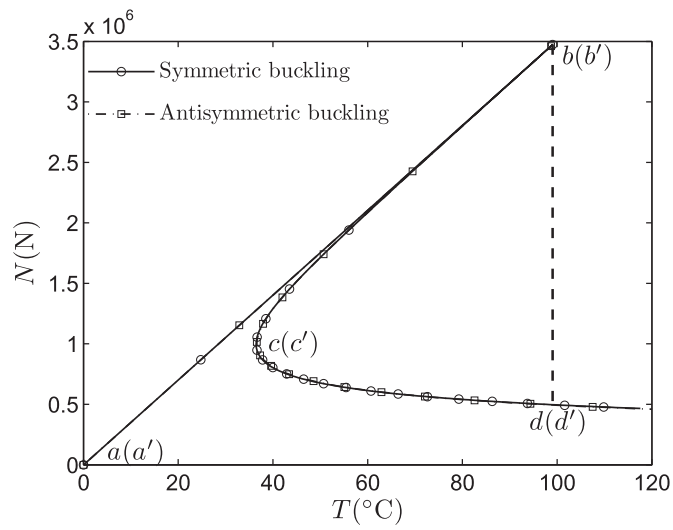


Fig. 7. Variation of axial force with temperature increase.

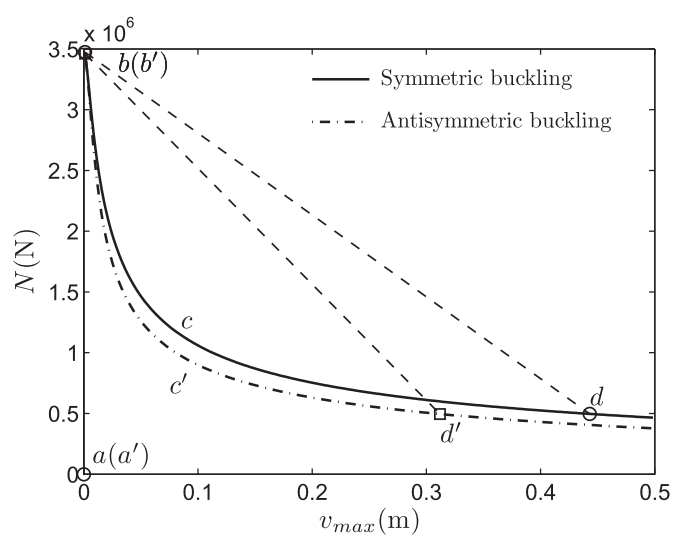
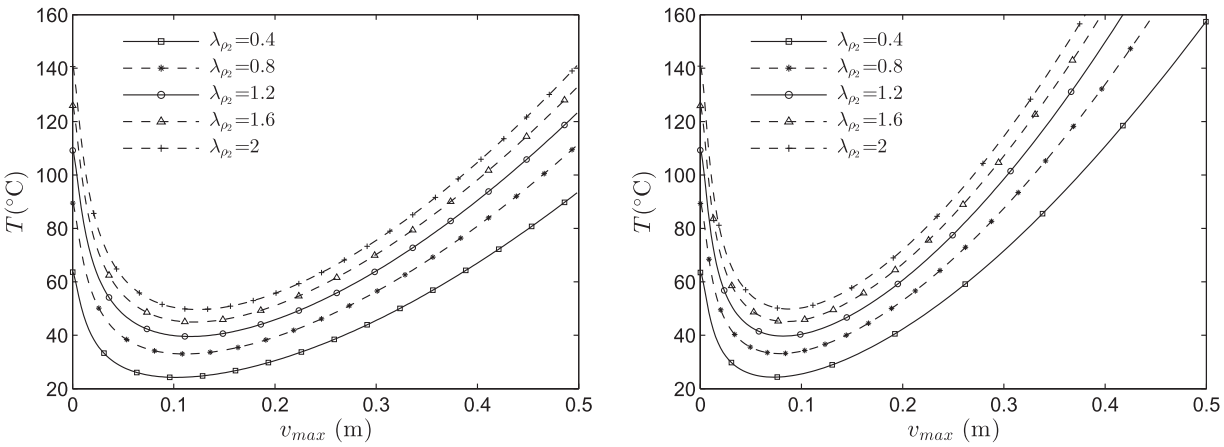


Fig. 8. Relationship between maximum lateral displacement and axial force.

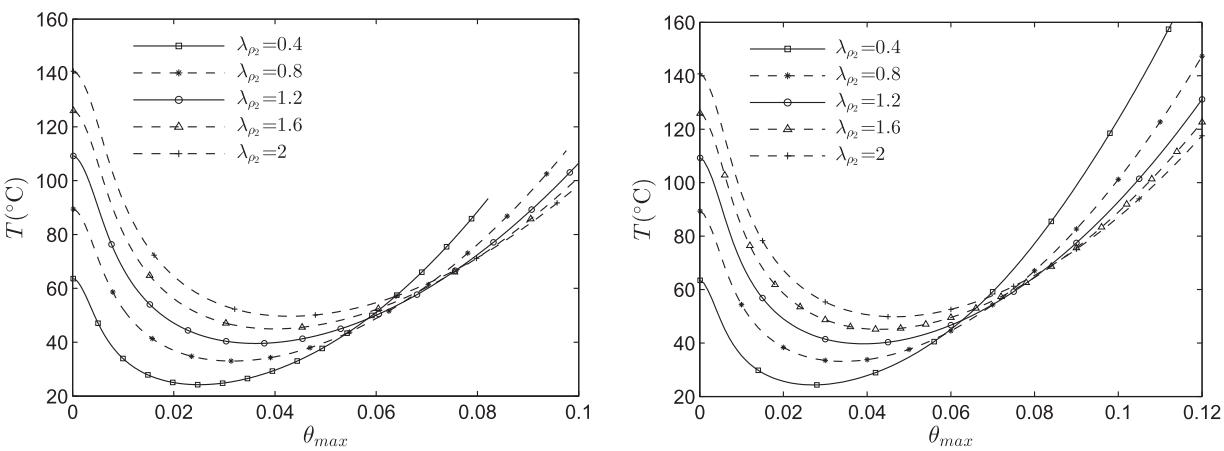
6.3. Influence of lateral resistance

The parameter ρ_2 represents the maximum lateral resistance of the ballast. By using a modified resistance parameter $\rho'_2 = \lambda\rho_2\rho_2$ (where $\lambda\rho_2 = 1$ corresponds to $\rho_2 = 3920$ N/m), the effects of the lateral resistance on the buckling are shown in Figs. 9–12. It can



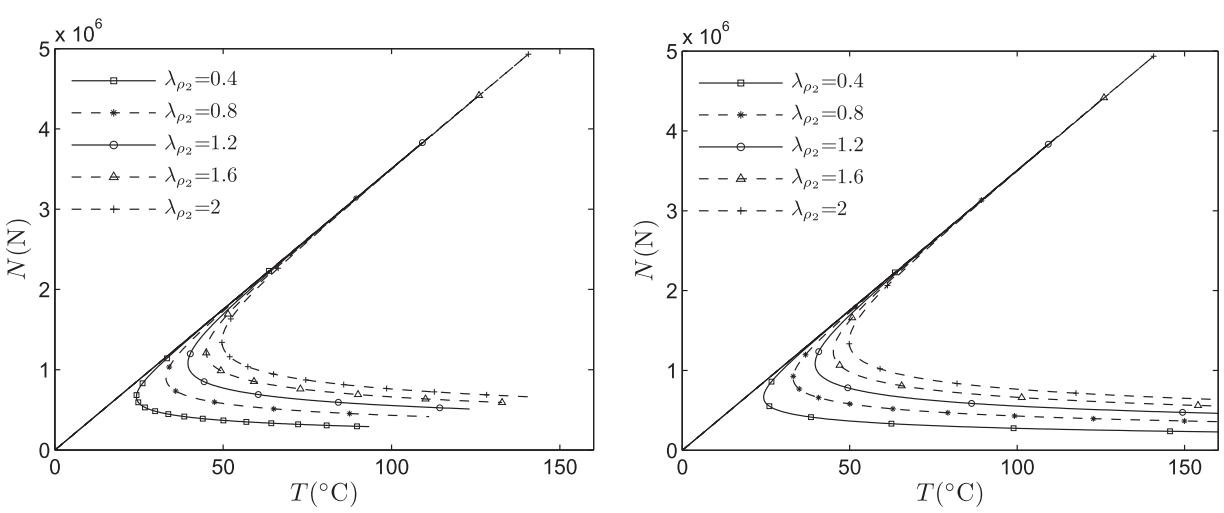
(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 9. Effects of lateral resistance (ρ_2) on postbuckling behaviour ($\upsilon_{max} - T$) of railway tracks.



(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 10. Effects of lateral resistance (ρ_2) on postbuckling behaviour ($\phi_{max} - T$) of railway tracks.



(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 11. Effects of lateral resistance (ρ_2) on postbuckling behaviour ($T - N$) of railway tracks.

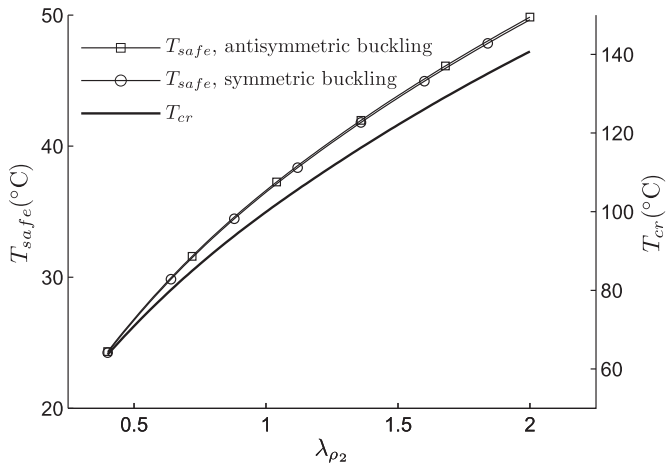


Fig. 12. Relationship between buckling temperature and lateral resistance.

be seen from Fig. 9 that larger lateral resistances lead to larger critical and safe temperatures. In the unstable postbuckling branch between the critical and safe temperatures, the displacements at a given temperature increase are larger for larger values of ρ_2, but following this the lateral displacements are larger for smaller values of ρ_2; this latter case being of interest as the remote stable branch is the equilibrium configuration to which the track snaps through. This behaviour is also observed for the maximum rotation of the track shown in Fig. 10. The relationships between the axial force and temperature presented in Fig. 11 show a similar trend, with the less stiff lateral restraint being associated with larger decreases of compression at the critical temperature. Fig. 12 plots the critical and safe temperatures as a function of the lateral restraint, and both of these increase with an increase of ρ_2. It can also be seen from this figure that the differences between the safe temperatures for symmetric and antisymmetric buckling are indistinguishable. The parameter η_2 (recast as $\eta'_2 = \lambda\eta_2\eta_2$ where $\lambda\eta_2 = 1$ corresponds to $\eta_2 = 300\ \mathrm{m}^{-1}$) can be seen in Figs. 13 and 14 to affect the critical temperature significantly, with lower values of η_2 decreasing the critical temperature significantly. However, the effect of η_2 on the safe temperature is only slight.

6.4. Influence of longitudinal resistance

The influence of the longitudinal resistance, as manifested by the parameters ρ_1 and η_1, has been investigated and the relationships between T and v_{max} and T and $\theta_{\max}$ are given in Figs. 15 and 16 respectively for values of ρ_1. It can be seen that this parameter has little effect on the postbuckling response. Counterpart relationships are given in Figs. 17 and 18 for values of the parameter η_1, which can be seen to have insignificant influence on the postbuckling response. Accordingly, simplifications of the buckling model may ignore the longitudinal resistance by comparison with the lateral ballast resistance.

6.5. Influence of imperfections

Imperfections in the railway system can be of many types, such as initial lateral displacements (geometric imperfection), weakening of the ballast (material imperfection), or the like. Perfectly straight railway tracks will remain straight under thermal loading and will bifurcate when a critical temperature is reached; tracks with an initial imperfection will deform non-linearly from the commencement of the thermal loading. Many types of imperfections have been considered (Lim et al., 2003; Tvergaard and Needleman, 1981). In this section, the shape of the lateral imperfection v_0 is assumed to be the same as that of the buckling modes, *viz.* symmetric or antisymmetric shapes, with a maximum value along the track of γ. The influence of these imperfections on the postbuckling are shown in Fig. 19, where it can be seen that non-linear behaviour takes place from the commencement of the thermal loading without bifurcation with an increase of the temperature. For railways with imperfections, the critical temperature can be considered sensibly as the peak (limit point) on the deformation *vs.* temperature curve. It can be seen that even with a small imperfection ($\gamma = 0.002$ m), there is a reduction in the critical temperature by as much as 40% compared with a straight track. However, when the amplitude is large (*e.g.* 0.015 m for symmetric buckling and 0.013 m for antisymmetric buckling), the deformations increase continuously without snap-through buckling with an increase in temperature characterised by so-called "progressive buckling". The imperfections can also be seen to increase the safe temperature to a minor extent, and the fluctuations in the equilibrium curve in Fig. 19(a) in the late postbuckling region are a result of the interaction of localisation and the symmetric initial imperfection acting against each other.

7. Conclusions

This paper has presented a study of the buckling and postbuckling response of railway tracks when subjected to thermal loading,

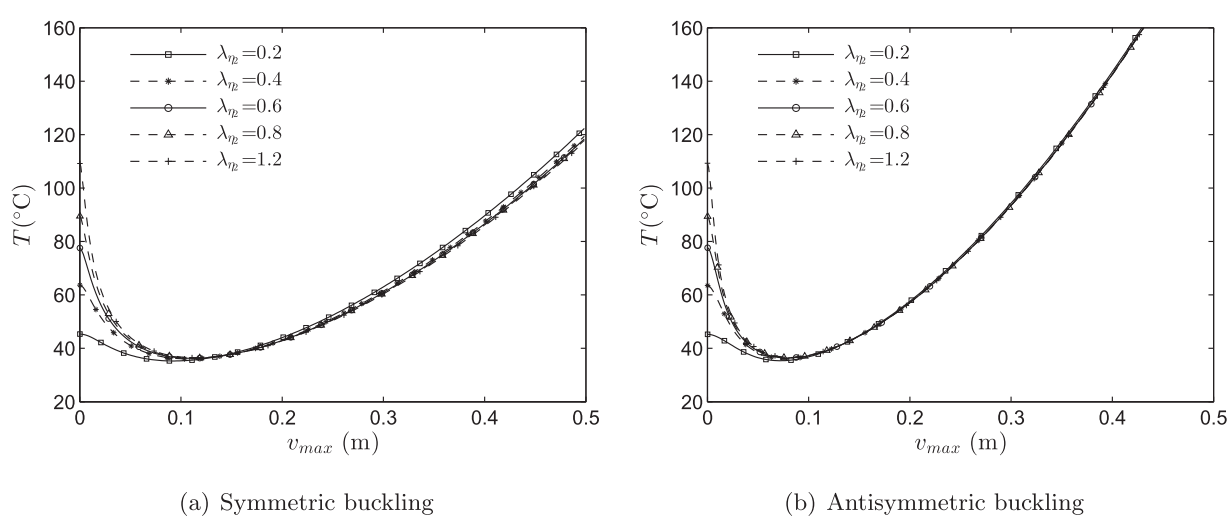
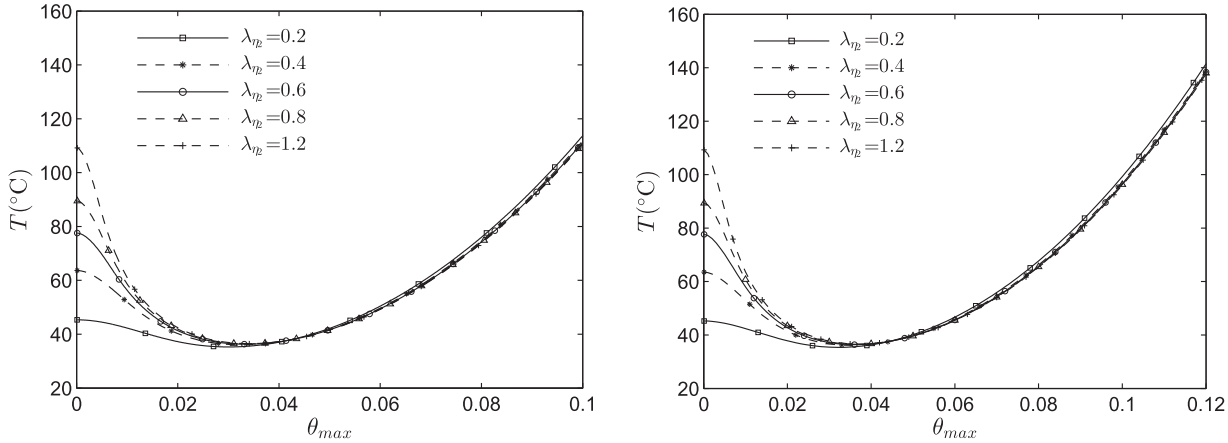
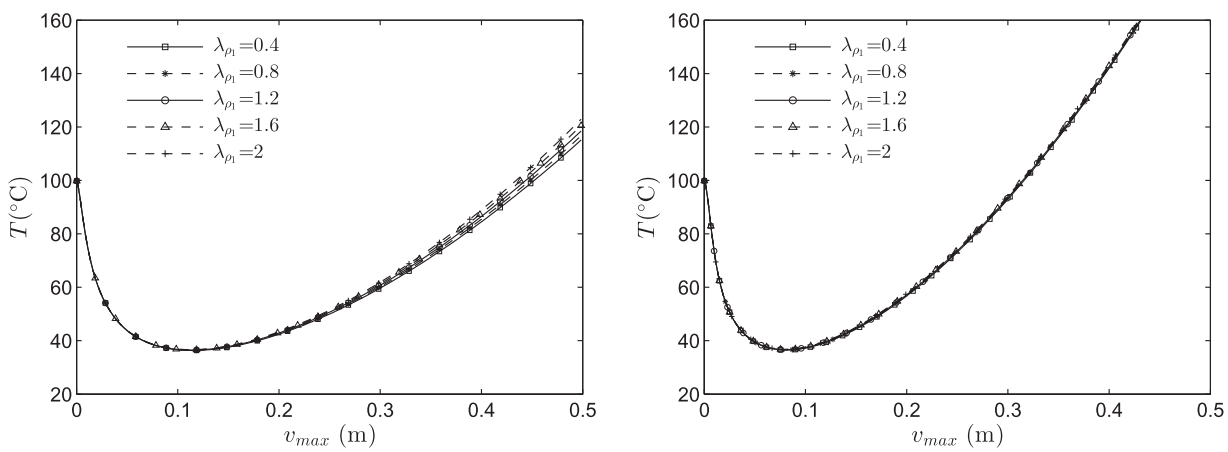


Fig. 13. Effects of lateral resistance (η_2) on postbuckling behaviour ($v_{max} - T$) of railway tracks.



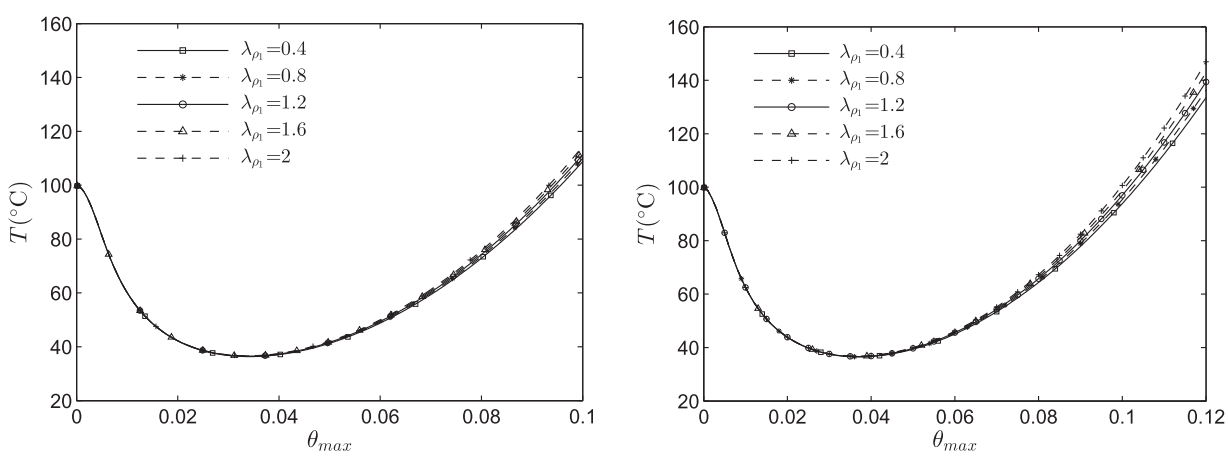
(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 14. Effects of lateral resistance (η_2) on postbuckling behaviour ($\phi_{max} - T$) of railway tracks.



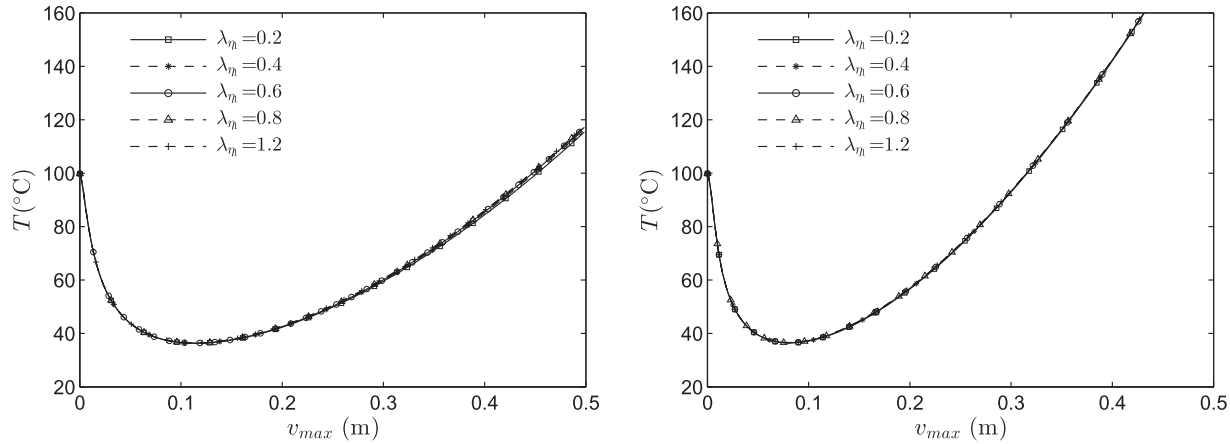
(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 15. Effects of longitudinal resistance (ρ_1) on postbuckling behaviour ($v_{max} - T$) of railway tracks.



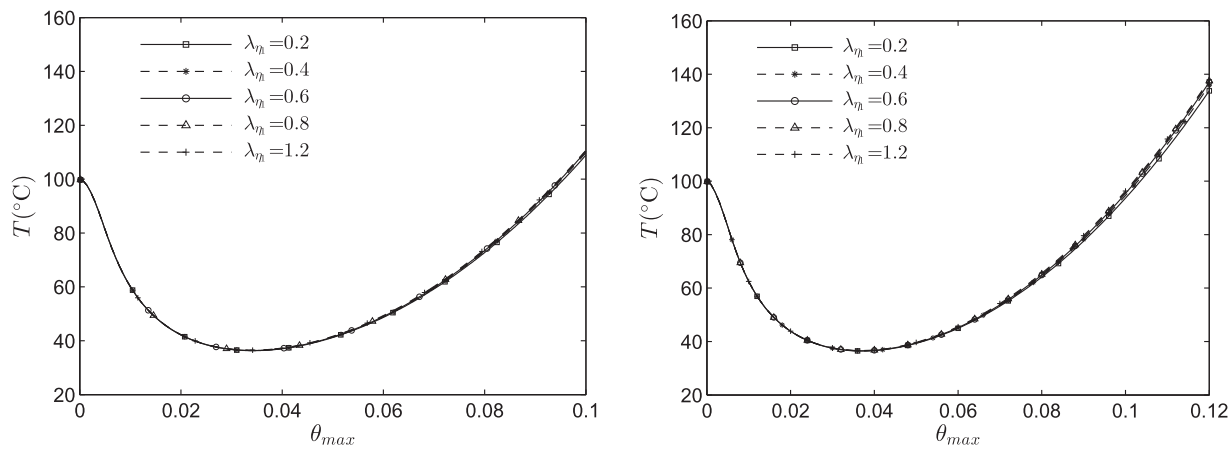
(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 16. Effects of longitudinal resistance (ρ_1) on postbuckling behaviour ($\phi_{max} - T$) of railway tracks.



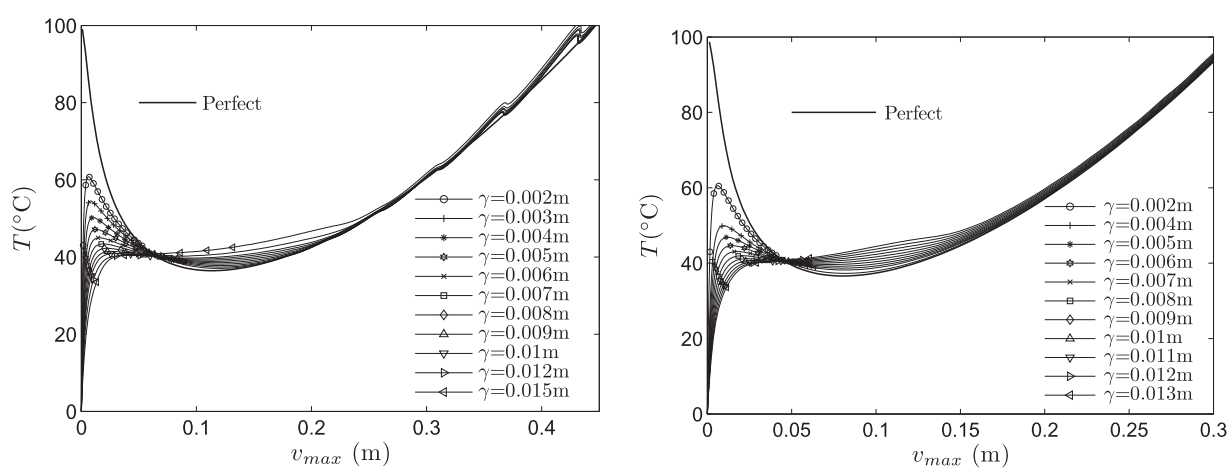
(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 17. Effects of longitudinal resistance (η_1) on postbuckling behaviour ($v_{max} - T$) of railway tracks.



(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 18. Effects of longitudinal resistance (η_1) on postbuckling behaviour ($\phi_{max} - T$) of railway tracks.



(a) Symmetric buckling (b) Antisymmetric buckling

Fig. 19. Effects of Imperfections on postbuckling behaviour of railway tracks.

including the effects of initial geometric imperfections. Using the principle of stationary total potential, the differential equations of buckling and postbuckling were derived, with single and multiple shooting techniques being used to arrive at the symmetric and antisymmetric postbuckling solutions numerically.

The study showed that for long railways, the first-order symmetric buckling mode plays the same role as the first-order antisymmetric one, and for perfectly-straight tracks, the initial postbuckling response is unstable and so the track snaps through to a remote stable equilibrium branch under an increase of temperature. It was also found that the postbuckling response is characterised by a localisation of the buckling mode with a significant decrease in the compression in the rail, as is observed in practice. The restraint provided by the ballast in the lateral direction was shown to be significant, whilst that in the longitudinal direction had little influence on the buckling. It was also shown that geometric imperfections affect the critical temperature significantly, and that for rails with large lateral imperfections the response under temperature loading is one of progressive buckling.

Acknowledgement

The work reported in this paper was supported by the Australian CSIRO Climate Adaptation Flagship through its Climate Adaptation Technology and Engineering for Extreme Events. This support is acknowledged with thanks.

Appendix

The terms in Eqs. (27) and (29) are

$$\Phi_1 = -v'^2,\ \Phi_2 = v'\left(u'+1\right),\ \Phi_3 = -4v'(u'^2v'' - 3u'u''v' - 2v'^2v'' + 2u'v'' - 3u''v' + v''),$$

$$\Phi_4 = -u'^3v'' - 7u'^2u''v - 13u'v'^2v'' + 5u''v'^3 - 3u'^2v'' - 14u'u''v' - 13v'^2v'' - 3u'v'' - 7u''v' - v'',$$

$$\Phi_5 = 3(u'v'' - u''v' + v'')(u'^3u''v'' + 7u'^2u''^2v' + 17u'u''v'^2v'' - u''^2v'^3 + 8v'^3v''^2 + 3u'^2u''v'' + 14u'u''^2v' + 17u''v'^2v'' + 3u'u''v'' + 7u''^2v' + u''v''),$$

$$\Phi_6 = -3u''u' - \tfrac{3}{2}u''u'^2 - u'' - \tfrac{1}{2}u''v'^2 + u''\alpha T - u'v'v'' - v'v'',$$

$$\Lambda_1 = -v'\left(u'+1\right),\ \Lambda_2 = \left(u'+1\right)^2,$$

$$\Lambda_3 = 3(u'v'' - u''v' + v'')(u'^3u''v'' + 7u'^2u''v' + 17u'u''v'^2v'' - u''^2v'^3 + 8v'^3v''^2 + 3u'^2u''v'' + 14u'u''^2v' + 17u''v'^2v'' + 3u'u''v'' + 7u''^2v' + u''v''),$$

$$\Lambda_4 = -4v'(u'^2v'' - 3u'u''v' - 2v'^2v'' + 2u'v'' - 3u''v' + v''),$$

$$\Lambda_5 = -u'^3v'' - 7u'^2u''v' - 13u'v'^2v'' + 5u''v'^3 - 3u'^2v'' - 14u'u''v' - 13v'^2v'' - 3u'v'' - 7u''v' - v'',$$

$$\Lambda_6 = -\tfrac{3}{2}v''v' - u'v'' - \tfrac{1}{2}u'^2v'' + v''\alpha T - u'u''v' - u''v',$$

while those in Eqs. (42) and (43) are

$$\bar{\Phi}_1 = -1 - \tfrac{1}{2}v'^2 - v'v'_0 + \alpha T,\ \bar{\Phi}_2 = -v''v'_0 - v'v'',$$

$$\bar{\Phi}_3 = -v'v'' - v''v'_0,$$

$$D_1 = \sqrt{v_0'^2 + 1},\ D_2 = \sqrt{v'^2 + 2v'v'_0 + v_0'^2 + 1},\ \bar{\Lambda}_1 = D_2^4,$$
$$\bar{\Lambda}_2 = -12v''\left(v' + v'_0\right)D_2^2,$$

$$\bar{\Lambda}_3 = 3v''\left(v'' - v''_0\right)\left(v'' + v''_0\right)\left(7v'^2 + 14v'v'_0 + 7v_0'^2 - 1\right),$$

$$\bar{\Lambda}_4 = 3v''v''_0\left(2v' + 2v'_0 + 1\right)\left(2v' + 2v'_0 - 1\right),$$

$$\bar{\Lambda}_5 = -\tfrac{1}{2}v''u'^2 - v''u' - \tfrac{3}{2}v''v'^2 - v''v_0'^2 + v''\alpha T - 3v''v'v'_0 - u'u''v'_0 - u''v' - u''v'_0 - u'u''v'.$$

References

Bradford, M.A., 2013. Thermal-induced pavement buckling in heatwaves. 5th International Conference on Climate Change – Impacts and Responses. Port Louis, Mauritius.
Chen, G., Baker, G., 2003. Rayleigh-Ritz analysis for localized buckling of a strut on a softening foundation by Hermite functions. Int. J. Solids Struct. 40, 7463–7474.
Dogneton, P., 1975. The experimental determination of the axial and lateral track--ballast resistance. In: Proceedings of the Symposium on Railroad Track Mechanics, pp. 171–196.
Donley, M.G., Kerr, A.D., 1987. Thermal buckling of curved railroad tracks. Int. J. Non-Linear Mech. 22, 175–192.
Esveld, C., 2001. Modern Railway Track, second ed. MRT Productions, Zaltbommel.
Hetenyi M. Beams on Elastic Foundation. Ann Abor: University of Michigan Press; 1974.
Hobbs, R.E., 1981. Pipeline buckling caused by axial loads. J. Constr. Steel Res. 2, 2–10.
Jeong D., Samavedam G., Kish A. Determination of Track Lateral Resistance from Lateral Pull Tests. Technical Report. Cambridge, MA: US Department of Transportation, Federal Railroad Administration; 1986.
Ju, G.T., Kyriakides, S., 1988. Thermal buckling of offshore pipelines. J. Offshore Mech. Arct. Eng., ASME 110, 335–364.
Keller, H.B., 1992. Numerical Methods for Two-point Boundary Value Problems. Dover Publications, New York.
Kerr, A.D., 1971. Lateral buckling of railroad tracks due to constrained thermal expansions – a critical survey. In: Proceedings of the Symposium on Railroad Track Mechanics. Pergamon Press, New York, pp. 141–169.
Kerr, A.D., 1978. Analysis of thermal track buckling in lateral plane. Acta Mech. 30, 17–50.
Kerr, A.D., 1980. An improved analysis for thermal track buckling. Int. J. Non-Linear Mech. 15, 99–114.
Kerr, A.D., Shade, P.J., 1984. Analysis of pavement blowups. Acta Mech. 52, 201–224.
Kish, A., Samavedam, G., 2013. Track Buckling Prevention: Theory, Safety Concepts, and Applications. US Department of Transportation, Federal Railroad Administration, Cambridge, MA.
Kish A., Samavedam G., Jeong D. Analysis of Thermal Buckling Tests on U.S. Railroads. Technical Report. Cambridge, MA: US Department of Transportation, Federal Railroad Administration; 1982.
Lai, W.M., Rubin, D., Krempl, E., 2010. Introduction to Continuum Mechanics, fourth ed. Elsevier, Boston.
Le Pen, L.M., Powrie, W., 2011. Contribution of base, crib, and shoulder ballast to the lateral sliding resistance of railway track: a geotechnical perspective. Proc. Inst. Mech. Eng. Part F 225, 113–128.
Li, S.-R., Batra, R.C., 2007. Thermal buckling and postbuckling of Euler-Bernoulli beams supported on nonlinear elastic foundations. AIAA J. 45, 712–720.
Lim, N.H., Han, S.Y., Han, T.H., Kang, Y.J., 2008. Parametric study on stability of continuous welded rail track – ballast resistance and track irregularity. Int. J. Steel Struct. 8, 171–181.
Lim, N.H., Park, N.H., Kang, Y.J., 2003. Stability of continuous welded rail track. Comput. Struct. 81, 2219–2236.
Love, A.E.H., 1944. A Treatise on the Mathematical Theory of Elasticity. Dover, Mineola NY.
Osborne, M.R., 1969. On shooting methods for boundary value problems. J. Math. Anal. Appl. 27, 417–425.
Peek, R., Yun, H.D., 2007. Floatation to trigger lateral buckles in pipelines on a flat seabed. J. Eng. Mech., ASCE 133, 442–451.
Pi, Y.-L., Bradford, M.A., Uy, B., 2007. A rational elasto-plastic spatially curved thin-walled beam element. Int. J. Numer. Methods Eng. 70, 253–290.
Pi, Y.-L., Trahair, N.S., 1994. Nonlinear inelastic analysis of steel beam-columns. I: Theory. J. Struct. Eng., ASCE 120, 2041–2061.
Samavedam G., Kanaan A., Pietrak J., Kish A., Sluz A. Wood Tie Track Resistance Characterization and Correlations. Technical Report. Cambridge, MA: US Department of Transportation, Federal Railroad Administration; 1986.
Samavedam G., Kish A., Jeong D. Parametric Studies on Lateral Stability of Welded Rail Track. Technical Report. Cambridge, MA: US Department of Transportation, Federal Railroad Administration; 1983.

Samavedam G., Kish A., Purple A., Schoengart J. Parametric Analysis and Safety Concepts of CWR Track Buckling. Technical Report. Cambridge, MA: US Department of Transportation, Federal Railroad Administration; 1993.
Simo, J.C., Vu-Quoc, L., 1991. A geometrically exact rod model incorporating shear and torsion-warping deformation. Int. J. Solids Struct. 27, 371–393.
Smith, S.T., Bradford, M.A., Oehlers, D.J., 2000. Unilateral buckling of elastically restrained rectangular mild steel plates. Comput. Mech. 26, 317–324.
Song, X., Li, S.R., 2007. Thermal buckling and post-buckling of pinned-fixed Euler Bernoulli beams on an elastic foundation. Mech. Res. Commun. 34, 164–171.
Stoer, J., Bulirsch, R., 2002. Introduction to Numerical Analysis, third ed. Springer Press, New York.
Timoshenko, S.P., Gere, J.M., 1961. Theory of Elastic Stability, second ed. McGraw Hill, New York.
Tvergaard, V., Needleman, A., 1981. On localized thermal track buckling. Int. J. Mech. Sci. 23, 577–587.
Van, M.A., 1997. PhD Thesis. Stability of continuous welded rail track. Delft: Delft University of Technology.
Wadee, M.K., Hunt, G.W., Whiting, A.I.M., 1997. Asymptotic and Rayleigh-Ritz routes to localized buckling solutions in an elastic instability problem. Proc. R. Soc. London, Ser. A 453, 2085–2107.
Wang, C.Y., 2010. Buckling of a weakened infinite beam on an elastic foundation. J. Eng. Mech., ASCE 136, 534–537.
Wu, B.S., Zhong, H.X., 1999. Postbuckling and imperfection sensitivity of fixed-end free-end struts on elastic foundation. Arch. Appl. Mech. 69, 491–498.
Yang, G., Bradford, M.A., 2015. Antisymmetric post-buckling localization of an infinite column on a nonlinear foundation with softening. Int. J. Struct. Stab. Dyn., 1540028.